Distribution of Matter in the Universe

From Lensing Clusters
to Large-Scale Structure

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The distribution of matter preserves a record of the composition, evolution, and structure of the Universe. This dissertation studies that distribution in two complementary regimes: gravitationally bound galaxy clusters and the cosmic large-scale structure.

Through strong and weak gravitational lensing, non-parametric mass reconstruction, lensing time delays, and models of the matter power spectrum, the work connects observations of individual clusters to the statistical description of cosmic structure. It examines mass–galaxy offsets, dark-matter substructure, baryonic effects on cosmological measurements, and the reconstructed mass maps of Hubble Frontier Field clusters.

The five research studies collected here show how lensing clusters and large-scale structure provide distinct but mutually reinforcing views of the matter content of the Universe.

COSMOLOGY GRAVITATIONAL LENSING DARK MATTER
LARGE-SCALE STRUCTURE GALAXY CLUSTERS

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